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Diffusion of Thought: Chain-of-Thought Reasoning in Diffusion Language Models

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Abstract

Recently, diffusion models have garnered significant interest in the field of text processing due to their many potential advantages compared to conventional autoregressive models. In this work, we propose Diffusion-of-Thought (DoT), a novel approach that integrates diffusion models with Chain-of-Thought, a well-established technique for improving the reasoning ability of autoregressive language models. In contrast to autoregressive language models that make decisions in a left-to-right, token-by-token manner, DoT allows reasoning steps to diffuse over time through a diffusion language model and offers greater flexibility in trading-off computation for reasoning performance. Our experimental results demonstrate the effectiveness of DoT in multi-digit multiplication, boolean logic, and grade school math problems, with a small diffusion model outperforming a much larger autoregressive model in both efficiency and accuracy. In addition to that, DoT showcases promising self-correction abilities and benefits from existing reasoning-enhancing techniques like self-consistency decoding. Our findings contribute to the understanding and development of reasoning with diffusion language models.

1 Introduction

Large language models (LLMs) have had a profound impact on the entire field of artificial intelligence [42, 50], transforming our approach to addressing classical problems in natural language processing and machine learning. Among the most notable aspects of LLMs is their remarkable reasoning ability, which many researchers consider to be a representative emergent capability brought about by LLMs [53]. Chain-of-thought prompting (CoT) [54]), which generates a series of intermediate reasoning steps in autoregressive (AR) way, has emerged as a central technique to support complex reasoning processes in LLMs. Despite advancements, errors in intermediate CoT steps can lead to inaccurate answers [32], posing self-correction difficulties [25], and concerns about CoT's inefficiency have been highlighted in recent studies [7].

Recently, diffusion models have attracted interest in text processing [33, 65, 69] as a result of success in the vision domain and distinctive modeling strengths over autoregressive models [34], offering potential benefits including global planning ability [59, 63], self correction [23] and efficiency [37]. As part of the research community effort, pre-trained diffusion language models such as Plaid [18] and SEDD [37] have shown significant progress in text generation capabilities. Although they have not yet attained the scale and capabilities of existing proprietary autoregressive LLMs like GPT-4 [42], these models have demonstrated performance on par with GPT2 [4] and the scaling law [27] in diffusion language models have been highlighted in Plaid. As a result, it becomes pertinent to explore the following question: *can diffusion language models also leverage the CoT-style technique to gain enhanced complex reasoning abilities?*

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This work presents a preliminary study on this question. We propose Diffusion of Thought (DoT), an inherent chain-of-thought method tailored for diffusion models. In essence, DoT progressively updates a sequence of latent variables representing thoughts in the hidden space, allowing reasoning steps to diffuse over time in parallel. We also introduce a multi-pass variant of DoT which focuses on generating one thought at a time to compensate for causal bias. To condition on complex queries, instead of using gradient-based classifier guidance [18, 33], DoT trains and samples from the denoising model using the classifier-free guidance as in Gong et al. [15], to provide more reliable controlling signals on exact tokens.

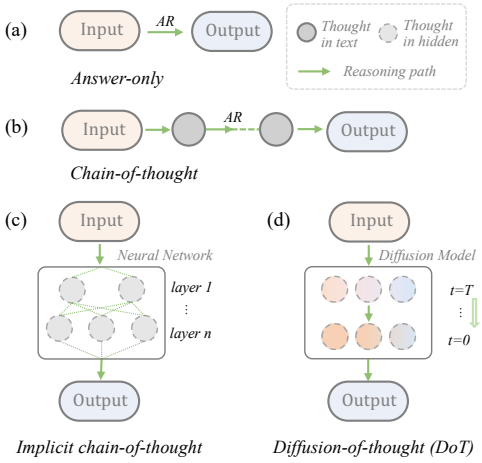

Figure 1: Illustration of reasoning approaches. (a) **Answer-only** and (b) **CoT** generate left-to-right tokens by prompting autoregressive language model. (c) **Implicit CoT** replaces horizontal reasoning (CoT) with vertical reasoning from shallow layer to deep layer [7]. (d) **DoT** generates reasoning path along with the diffusion timesteps.

Furthermore, to improve the self-correcting capability of the diffusion model, DoT integrates training-time sampling algorithms to learn to recover from errors originating from prior or current reasoning steps. This feature offers a fresh angle on the issue of error accumulation [25, 32] inherent in autoregressive models. Finally, we adapt a conditional ODE Solver [39] for DoT during inference time to accelerate the inference of continuous diffusion models. We show DoT enjoys flexibility in trading off computation (reasoning time) and performance as more complex problems may necessitate increased computation in reasoning [2, 54].

From a methodological standpoint, DoT shares similarities with the recently proposed Implicit CoT approach [7], where the latter learns thoughts in hidden states across transformer layers to improve the time efficiency of autoregressive CoT generation. A schematic illustration of CoT, Implicit CoT, and DoT can be found in Figure 1.

The main contributions of our paper are threefold:

1. We first introduce the reasoning technique for diffusion models (DoT), and showcase its advantages in simple reasoning tasks (digit multiplication and boolean logic) when compared to autoregressive CoT and Implicit CoT. DoT achieves up to $27\times$ speed-up without performance drop (§4.2).
2. We further adapt DoT to continuous and discrete diffusion base models, and introduce two training-time sampling algorithms to improve its self-correction ability. DoT exhibits superior performance compared to GPT2 with CoT on grade school math problems, enabling a small diffusion model to outperform a 4.6x larger autoregressive model, showing the potential of text diffusion models for complex reasoning (§4.3).
3. Our analysis demonstrates the flexibility of DoT in the trade-off between reasoning time and performance (§4.4), and showcases DoT's self-correction capability (§4.6). We also find that self-consistency decoding can further improve DoT and its multi-pass variant (§4.5).

Although it is challenging for current pre-trained diffusion language models to directly compete with LLMs that are hundreds of times larger in parameter size, our study emphasizes the possibility of their complex reasoning abilities and highlights the substantial potential in developing LLMs that go beyond the autoregressive paradigm. We release all the codes at https://github.com/HKUNLP/diffusion-of-thoughts.

2 Preliminaries

This section introduces key concepts and notations in diffusion models for text generation. Detailed formulations and derivations are provided in Appendix A.

A typical diffusion model contains the forward and reverse process. For each forward step $q(\mathbf{z}_t|\mathbf{z}_{t-1})$, we gradually inject noise into the data representation $\mathbf{z}_{t-1}$ from the last timestep to obtain $\mathbf{z}_t$. Here $t = 1, 2, ..., T$ and the larger t corresponds to noisier data. For reverse process, the ultimate goal is to recover the original $\mathbf{z}_0$ by denoising $\mathbf{z}_t$: $p_\theta(\mathbf{z}_{0:T}) := p(\mathbf{z}_T) \prod_{t=1}^{T} p_\theta(\mathbf{z}_{t-1}|\mathbf{z}_t)$. We model the learning process $p_\theta(\mathbf{z}_{t-1}|\mathbf{z}_t)$ using the proposed diffusion model $\mathbf{z}_\theta(\mathbf{z}_t, t)$.

Previous text generation using diffusion models almost contains two categories: (1) Continuous diffusion models such as Diffusion-LM [33], which relies on a mapping function between the real values and feasible integral point; (2) Discrete diffusion models like D3PM [1], which directly formulate the problem as the integer program. Continuous diffusion models map the discrete text $\mathbf{w}$ into a continuous space through an embedding function $\textsc{Emb}(\mathbf{w})$, and its inverse operation is called rounding. The forward perturbations are applied according to $q(\mathbf{z}_t|\mathbf{z}_{t-1}) = \mathcal{N}(\mathbf{z}_t; \sqrt{1-\beta_t}\mathbf{z}_{t-1}, \beta_t \mathbf{I})$, where $\beta_t \in (0, 1)$ represents different scales of the Gaussian noise. Plaid [18] is a continuous diffusion language model trained from scratch on 314B tokens with 1024 context size. It is currently the largest scale diffusion language model with 1.3B parameters. In the case of discrete diffusion models, each $\mathbf{z}_t$ is represented as a discrete random variable using one-hot vectors in $\{0, 1\}^K$, where K denotes the vocabulary size. They define $q(\mathbf{z}_t|\mathbf{z}_{t-1})$ through a transition matrix, making it a point mass with probability on an absorbing state `[MASK]` or a uniform distribution over the vocabulary size. SEDD [37] is a recently trained-from-scratch discrete diffusion language model with small and medium size similar to GPT2.

For sequence-to-sequence (seq2seq) generation, which involves a pair of sequences $\mathbf{w}^x$ and $\mathbf{w}^y$, DiffuSeq [15] treats these two sequences as a single one $\mathbf{w}^z = \mathbf{w}^{[x;y]}$ and uses a left-aligned mask $[\mathbf{0}; \mathbf{1}]$ during the forward and reverse diffusion process to distinguish them. Unlike traditional diffusion models that corrupt the entire $\mathbf{z}_t$, DiffuSeq only adds noise to those entries with the mask value of 1 (e.g., $\mathbf{y}_t$). This modification, termed partial noising, tailors diffusion models for conditional language generation, and set a difference between the gradient-based token guidance in [33] and [18].

3 Diffusion-of-Thoughts

In this section, we begin with an overview of our method and its relationship with other reasoning paradigms (§3.1). We then introduce Diffusion-of-Thoughts (DoT) as well as its multi-pass variant (DoT^{MP}; §3.2), as illustrated in Figure 2. Following this, we outline the implementation of our training (§3.3) and inference (§3.4) protocols.

3.1 Overview

Without loss of generality, we use the mathematical problem-solving task as our running example. A problem statement and its correct answer are denoted as $\mathbf{s}$ and $\mathbf{a}$, respectively. We employ a language model with parameters θ, represented as p_θ^{LM}, to find the solution for each problem. For regular usage of language models without Chain-of-Thoughts (CoT), the final answer $\mathbf{a}$ is generated directly as $\mathbf{a} \sim p_\theta^{LM}(\mathbf{a}|\mathbf{s})$. The CoT approach introduces meaningful intermediate steps or rationales $\mathbf{r}_1, \ldots, \mathbf{r}_n$ for language models to bridge $\mathbf{s}$ and $\mathbf{a}$, resulting in the output $\mathbf{a} \sim p_\theta^{LM}(\mathbf{a}|\mathbf{s}, \mathbf{r}_{1\ldots n})$. For implicit CoT [7], the hidden representations of rationales $\mathbf{z}_1, \ldots, \mathbf{z}_n$ are distilled into transformer layers, leading to $\mathbf{a} \sim p_\theta^{iCoT}(\mathbf{a}|\mathbf{s}, \mathbf{z}_{1\ldots n})$. Similarly but differently, for DoT, these representations are distributed over diffusion timestep t as $\mathbf{a} \sim p_\theta^{DoT}(\mathbf{a}|\mathbf{s}, \mathbf{z}_t)$, where $\mathbf{z}_t$ corresponds exactly to the noised data in diffusion models.

3.2 Modeling

We begin by observing the gradient-based token guidance fails to do accurate conditioning as the model cannot exactly recover each conditioning token (see Table 2). This is vital, especially in mathematical reasoning, as it is expected to perform reasoning based on exact tokens (e.g., numbers) in the problem statement, rather than more compact gradient signals. For this, we adopt DiffuSeq-style [15] classifier-free conditioning during the fine-tuning of Plaid, where all rationales are generated by the backward diffusion process in parallel, with all the conditional tokens fixed as still. Specifically, the problem context s is concatenated with the rationales $\mathbf{r}_{1\ldots n}$ during training and sampling, while the noise is only partially imposed to the rationale part in $\mathbf{r}_{1\ldots n}$, keeping $\mathbf{s}$ anchored as the condition.